MANNERU HITESH

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PROFESSIONAL SUMMARY

Third-year CSE undergraduate at GITAM University (CGPA: 8.78) with hands-on experience building data-driven systems using Python, FastAPI, and SQL. Developed full-stack applications demonstrating applied software engineering, RESTful API design, and structured data processing — skills directly transferable to ML pipelines and feature engineering. Advancing into machine learning through self-directed study of supervised learning, probability, and linear algebra, with active project work in intelligent systems and conversational AI. Seeking Amazon ML Summer School 2026 to bridge strong engineering foundations with rigorous ML training.

EDUCATION

Bachelor of Technology — Computer Science Engineering | *GITAM University, VisakhapatnamAug 2024 – Apr 2028*

CGPA: 8.78 • *Currently Pursuing (Second Year → Third Year)*

Relevant Coursework: Data Structures & Algorithms • Object-Oriented Programming • Database Management Systems • Software Engineering • Web Technologies • Python • Java • C Programming

Self-Study (ML-Relevant): Supervised Learning • Probability & Statistics • Linear Algebra • Model Evaluation • NumPy • Pandas

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, TypeScript
ML / Data: NumPy, Pandas, Scikit-learn (learning), SQL
Frameworks: FastAPI, Flask, Spring Boot, REST APIs
Databases: MySQL, PostgreSQL, SQLite
Tools & Platforms: Git, VS Code, IntelliJ, Eclipse
Core Areas: DSA, OOP, System Design, API Development

PROJECTS

Smart Irrigation System — Sensor-Driven Automation | *Academic Project2025*

- Designed an end-to-end IoT system that ingested real-time soil moisture and flow sensor data to automate irrigation decisions, reducing manual intervention entirely.
- Implemented threshold-based decision logic to detect pipe leakages via flow sensor anomalies — a rule-based classification approach analogous to supervised ML inference.
- Integrated software and hardware components into a unified pipeline: data acquisition → processing → automated actuation — directly mirroring ML data pipelines.
- Technologies: Python, Sensor Integration, Automation Logic, System Design

Project N — Full-Stack Streaming Platform (Netflix Clone) | *Academic Project2024–2025*

- Built 8+ RESTful API endpoints using FastAPI (Python) for user management and content retrieval, handling structured data ingestion and query-based filtering.
- Implemented JWT-based authentication securing user sessions, applying token validation logic across all protected routes.
- Designed relational database schema for users and content metadata in SQL, enforcing data integrity — a foundation directly applicable to ML feature stores and structured datasets.
- Delivered a responsive full-stack UI with modular backend architecture enabling clean separation of data, logic, and presentation layers.
- Technologies: Python, FastAPI, HTML, CSS, JavaScript, SQL, JWT, Git

Leo — AI-Powered Virtual Assistant | *Personal Project2025 – Ongoing*

- Building a conversational AI assistant to study natural language processing, intent classification, and human-computer interaction patterns.
- Exploring dialogue management and context retention mechanisms as foundations for understanding sequence modelling and language model behaviour.
- Technologies: Python, NLP concepts, Conversational AI
- (Starting stage,Still working on this)

Assessment Management System | *Academic Project2024*

- Developed a web application automating quiz delivery, response collection, and performance scoring — generating structured output datasets amenable to further analysis.
- Applied modular Python architecture and SQLite for persistent data storage, enabling repeatable data collection across test sessions.
- Technologies: Python, Flask, SQLite, HTML, CSS, JavaScript

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- **Advanced to Round 2 of SIH 2024 — collaborated on rapid prototyping and data-driven solution design under real-world problem constraints with a cross-functional team.** Smart India Hackathon 2024:
- **Solved 128+ algorithmic problems spanning dynamic programming, graphs, trees, and recursion — building computational thinking relevant to ML optimisation and algorithm design.** LeetCode:
- **Consistent practice on HackerRank strengthening problem decomposition, time complexity analysis, and implementation of core computer science fundamentals.** Coding Practice:

AVAILABILITY

Available for full programme duration of Amazon ML Summer School 2026. Open to relocating for in-person sessions.